

DSG-CTTA: Continual Test-Time Adaptation Discovery Report

Rigorous Empirical Characterization: Performance Disparities, Subgroup Dynamics, and Prequential Robustness Across Continual Adaptation Baselines

Protocol: v1.0.0-canonical	Partition: final_test.csv (60 utts, 6 spks)	Model: wav2vec2-base-960h
Paradigm: Prequential ($B_t \rightarrow \theta_t$)	Isolation: Air-Gapped Audio (0 text / 0 meta)	Scope: 4 Methods $\times$ 3 Orders (12 runs)

1. Executive Summary & Research Logic

Stage 3 is technically successful as a CTTA discovery pilot, but it is not strong enough to conclude that adaptation-induced disparity does not occur. The evaluation pipeline establishes a mathematically verified prequential streaming runtime across adaptation methods (SUTA, DSUTA, DMSUTA). We approve moving forward to **Stage 4: Larger-Scale Mechanism / Phenomenon Characterization**, but we explicitly **do NOT approve moving directly to Stage 5 (DSG)**. The static Stage 2 disparity baseline (D_0) serves strictly as the pre-adaptation reference (θ_0), not as evidence that DSG is necessary.

BENCHMARK SCALE & WORD-LEVEL SENSITIVITY: The final_test stream contains 60 utterances, 6 speakers, and 552 reference words (~92 words per group). On a 92-word subgroup, **a single word classification alteration represents $\Delta WER_g = 1/92 \approx 1.09\%$** . The observed reverse-order regression of +1.09% corresponds literally to **one word**. A +1.09% change is a real observed numerical shift, but cannot be claimed as practically meaningful or statistically negligible without confidence intervals and larger-scale multi-speaker evaluation.

Prequential Protocol: $B_t \rightarrow \theta_t$ Prediction $\rightarrow$ Offline Scoring $\rightarrow$ Unlabeled Adaptation $\rightarrow \theta_{t+1}$. Window B_t is never evaluated using θ_{t+1} . Online runtime receives zero reference text, labels, speaker IDs, or accent metadata. Passing 25/25 automated tests proves implementation obeys invariants, **not** that the sample size is powered for a null claim.

Speech Variety vs Accent: Groups represent *L1-associated English speech varieties* where phonology, proficiency, and acoustics are intertwined.

Window Size Protocol (K=4): Chosen as the pre-registered default in configs/default_config.yaml (60/4 = 15 windows). In Stage 4, $K \in \{1, 4, 5, 10\}$ will be calibrated on development/calibration splits before untouched final evaluation.

Table 1: Canonical Stream Results (ORDER_A, K=4) Without Safety Intervention

Method	Live Corpus WER	Δ_R (Overall)	Disparity D	Δ_D (Disparity)	Max Regr $\max_g \Delta_g$	Worst Subgroup	Updates / Resets
No-Adapt (Control)	85.51%	0.00%	20.65%	0.00%	0.00%	None	0 / 0
SUTA	85.14%	-0.37%	19.57%	-1.08%	0.00%	None	15 / 0
DSUTA	85.33%	-0.18%	19.57%	-1.08%	0.00%	None	15 / 1
DMSUTA	85.51%	0.00%	20.65%	0.00%	0.00%	None	15 / 0

Table 2: L1 Speech Variety Performance Breakdown on Canonical ORDER_A

L1 Variety	No-Adapt WER	SUTA WER	SUTA Δ_g	DSUTA WER	DSUTA Δ_g	DMSUTA WER	DMSUTA Δ_g
Arabic	89.13%	89.13%	0.00%	89.13%	0.00%	89.13%	0.00%
Hindi	97.83%	96.74%	-1.09%	96.74%	-1.09%	97.83%	0.00%
Korean	82.61%	82.61%	0.00%	82.61%	0.00%	82.61%	0.00%
Mandarin	77.17%	77.17%	0.00%	77.17%	0.00%	77.17%	0.00%
Spanish	85.87%	85.87%	0.00%	85.87%	0.00%	85.87%	0.00%
Vietnamese	80.43%	79.35%	-1.08%	80.43%	0.00%	80.43%	0.00%

2. Prequential Trajectories & Full Multi-Order Matrix

Four high-resolution prequential trajectory figures capture adaptation progression, disparity dynamics, and group trajectories:



Table 3: Full Multi-Order Robustness Matrix (4 Methods × 3 Stream Orders)

To test stream-order sensitivity, No-Adapt, SUTA, DSUTA, and DMSUTA were evaluated across three stream permutations: ORDER_A (Canonical), ORDER_B (Reverse), and ORDER_C (Permuted):

Stream Order	Method	Corpus WER	Δ_R	Disparity D	Δ_D	Max Regr $\max_g \Delta_g$	Worst Regressed Subgroup
ORDER_A	No-Adapt	85.51%	0.00%	20.65%	0.00%	0.00%	None
ORDER_A	SUTA	85.14%	-0.37%	19.57%	-1.08%	0.00%	None
ORDER_A	DSUTA	85.33%	-0.18%	19.57%	-1.08%	0.00%	None
ORDER_A	DMSUTA	85.51%	0.00%	20.65%	0.00%	0.00%	None
ORDER_B	No-Adapt	85.51%	0.00%	20.65%	0.00%	0.00%	None
ORDER_B	SUTA	85.69%	+0.18%	20.65%	0.00%	+1.09%	Arabic
ORDER_B	DSUTA	85.14%	-0.37%	19.57%	-1.08%	+1.09%	Vietnamese
ORDER_B	DMSUTA	85.51%	0.00%	19.57%	-1.08%	+1.09%	Arabic
ORDER_C	No-Adapt	85.51%	0.00%	20.65%	0.00%	0.00%	None
ORDER_C	SUTA	85.51%	0.00%	20.65%	0.00%	+1.09%	Spanish
ORDER_C	DSUTA	85.14%	-0.37%	20.65%	0.00%	0.00%	None
ORDER_C	DMSUTA	85.33%	-0.18%	20.65%	0.00%	+1.09%	Vietnamese

3. Statistical Uncertainty & Evidence Boundary

Table 4: Paired Speaker-Cluster Bootstrap Analysis (B=1,000 resamples, $\alpha=0.05$). Confidence intervals explicitly distinguish observed numerical changes from statistically verified effects:

Order	Method	Metric	Point Est	Boot Mean	Std Error	95% Confidence Interval	95% UCB
ORDER_A	SUTA	Δ_R	-0.37%	-0.35%	0.20%	[-0.72%, 0.00%]	0.00%
ORDER_A	SUTA	Δ_D	-1.08%	-0.72%	0.52%	[-1.09%, 0.00%]	0.00%
ORDER_A	SUTA	$\max_g \Delta_g$	0.00%	0.00%	0.00%	[0.00%, 0.00%]	0.00%
ORDER_A	DSUTA	Δ_R	-0.18%	-0.17%	0.16%	[-0.54%, 0.00%]	0.00%
ORDER_A	DSUTA	Δ_D	-1.08%	-0.71%	0.52%	[-1.09%, 0.00%]	0.00%
ORDER_A	DSUTA	$\max_g \Delta_g$	0.00%	0.00%	0.00%	[0.00%, 0.00%]	0.00%
ORDER_A	DMSUTA	Δ_R	0.00%	0.00%	0.00%	[0.00%, 0.00%]	0.00%
ORDER_A	DMSUTA	Δ_D	0.00%	0.00%	0.00%	[0.00%, 0.00%]	0.00%
ORDER_A	DMSUTA	$\max_g \Delta_g$	0.00%	0.00%	0.00%	[0.00%, 0.00%]	0.00%
ORDER_B	SUTA	Δ_R	+0.18%	+0.18%	0.16%	[0.00%, +0.54%]	+0.54%
ORDER_B	SUTA	Δ_D	0.00%	+0.29%	0.48%	[0.00%, +1.09%]	+1.09%
ORDER_B	SUTA	$\max_g \Delta_g$	+1.09%	+0.74%	0.51%	[0.00%, +1.09%]	+1.09%
ORDER_B	DSUTA	Δ_R	-0.37%	-0.37%	0.32%	[-0.91%, +0.36%]	+0.18%
ORDER_B	DSUTA	Δ_D	-1.08%	-1.07%	0.16%	[-1.09%, -1.09%]	-1.09%
ORDER_B	DSUTA	$\max_g \Delta_g$	+1.09%	+0.73%	0.51%	[0.00%, +1.09%]	+1.09%
ORDER_B	DMSUTA	Δ_R	0.00%	+0.01%	0.25%	[-0.54%, +0.54%]	+0.36%
ORDER_B	DMSUTA	Δ_D	-1.08%	-0.42%	0.96%	[-1.09%, +1.09%]	+1.09%
ORDER_B	DMSUTA	$\max_g \Delta_g$	+1.09%	+0.74%	0.51%	[0.00%, +1.09%]	+1.09%
ORDER_C	SUTA	Δ_R	0.00%	0.00%	0.25%	[-0.54%, +0.54%]	+0.36%
ORDER_C	SUTA	Δ_D	0.00%	-0.22%	0.58%	[-1.09%, +1.09%]	+1.09%
ORDER_C	SUTA	$\max_g \Delta_g$	+1.09%	+0.74%	0.51%	[0.00%, +1.09%]	+1.09%
ORDER_C	DSUTA	Δ_R	-0.37%	-0.37%	0.21%	[-0.72%, 0.00%]	0.00%
ORDER_C	DSUTA	Δ_D	0.00%	-0.36%	0.51%	[-1.09%, 0.00%]	0.00%
ORDER_C	DSUTA	$\max_g \Delta_g$	0.00%	0.00%	0.00%	[0.00%, 0.00%]	0.00%
ORDER_C	DMSUTA	Δ_R	-0.18%	-0.19%	0.39%	[-1.09%, +0.54%]	+0.36%
ORDER_C	DMSUTA	Δ_D	0.00%	-0.58%	0.96%	[-2.17%, 0.00%]	0.00%
ORDER_C	DMSUTA	$\max_g \Delta_g$	+1.09%	+0.73%	0.51%	[0.00%, +1.09%]	+1.09%

Scientific Boundaries: What Stage 3 Has vs Has NOT Demonstrated

NOT DEMONSTRATED BY CURRENT EVIDENCE:	DEMONSTRATED & EMPIRICALLY OBSERVED:
× CTTA causes disparity amplification ($\Delta_D > \delta_D$) × CTTA causes systematic subgroup regression ($\&\max_g \Delta_g > \delta_G$) × SUTA is harmful / DSUTA is harmful × DMSUTA is safe × DSG is necessary × DSG will improve fairness	✓ CTTA can slightly change overall WER ($\Delta_R \in [-0.37\%, +0.18\%]$) ✓ SUTA / DSUTA produced small gains on ORDER_A (-0.37%, -0.18%) ✓ DMSUTA produced flat aggregate WER on ORDER_A (0.00%) ✓ Small order-dependent subgroup regression (+1.09%, 1 word) observed ✓ No disparity amplification observed on any tested stream ($\Delta_D \leq 0.00\%$) ✓ Prequential/label-isolation passed all 25 automated tests

4. Coupling Determination, Stage 4 Mandate & Provenance

CANONICAL SCIENTIFIC DETERMINATION (MANDATORY STANDARD):
"Meaningful coupling is not demonstrated on the current benchmark stream. The observed changes are small, and the current sample size and unfrozen practical-effect threshold prevent a stronger conclusion regarding the absence of adaptation-induced subgroup harm."

- *Practical Significance Threshold (δ):* Status remains UNKNOWN: delta not frozen sufficiently for final discovery. No post-hoc threshold was manufactured from test data.
- *Permissible Scientific Claim:* No practically meaningful coupling was demonstrated under current pilot conditions.
- *Impermissible Claim:* We do NOT claim proof that CTTA does not cause disparity amplification.

Stage 4 Mandate & Execution Architecture

1. NO-GO for Stage 5 DSG on Current Final Test Stream: Proceeding directly to build DSG safety controllers without empirical evidence of disparity amplification would violate scientific integrity.
2. PROCEED to Stage 4 (Phenomenon Characterization) to test boundary conditions:
 - **Stage 4A (Data-Scale Expansion):** Maximize disjoint independent speakers across Dev, Cal, Sentinel, and Final Test splits.
 - **Stage 4B (Freeze δ_G / δ_D):** Statistically derive δ via calibration power analysis and smallest practically important effect.
 - **Stage 4C (Window-Size Experiments):** Sweep $K \in \{1, 4, 5, 10\}$ on Dev/Cal splits before untouched final evaluation.
 - **Stage 4D (Multi-Order Matrix):** Test all primary CTTA methods across multiple stream permutations.
 - **Stage 4E (Acoustic Stress Testing):** Introduce controlled distribution shifts (additive babble noise, SNR steps, non-stationary shift).
 - **Stage 4F (Speaker-Cluster Uncertainty):** Full paired bootstrap 95% CIs for Δ_R , Δ_g , and Δ_D across all conditions.
 - **Stage 4G (Condition-Wise Boundaries):** Map multidimensional instability boundaries (Method $\times$ Order $\times$ K $\times$ Shift Severity).
 - **Stage 4H (GO / NO-GO for DSG):** Implement DSG in Stage 5 *only* if coupling is demonstrated under characterized conditions.

Technical Provenance & Verification Audit

Artifact / Verification Check	Observed Fingerprint / Value	Verification Status
Stream ORDER_A Hash	e02b45a1fd007011f1816e88907de3507d4b29bb883c58b4...	VERIFIED (Deterministic)
Stream ORDER_B Hash	e45bd9d0921c06db18d80f68e09f5fa414a383b0fce04bda...	VERIFIED (Distinct)
Stream ORDER_C Hash	5dc02804e713625afae2cfbbd8e03e7d446b0b2e3e6015ca...	VERIFIED (Permuted)
Prequential Live Invariant	Window B_t scored with theta_t strictly prior to update	VERIFIED (Air-Gapped)
Multi-Order Matrix Scope	All 4 Methods x 3 Stream Orders (12 experimental runs)	VERIFIED (Complete)
Speaker-Cluster Bootstrap	B=1,000 paired resamples for Delta_R, Delta_D, max Delta_g	VERIFIED (Alpha=0.05)
Automated Test Suite	25 / 25 Passing (19 baseline + 6 prequential)	PASS (100%)